0-1 knapsack problem

```c
#include<stdio.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

void knapsack(int n1,int n2,int profit[],int weight[])
{
    int selected[100]={0};
    int v[n1+1][n2+1];
    for(int i=0;i<=n1;i++)
    {
        v[i][0]=0;
    }
    for(int j=0;j<=n2;j++)
    {
        v[0][j]=0;
    }
    for(int i=1;i<=n1;i++)
    {
        for(int j=1;j<=n2;j++)
        {
            if(weight[i]>j)
            {
                v[i][j]=v[i-1][j];
            }
            else
            {
                v[i][j]=max(v[i-1][j],v[i-1][j-weight[i]]+profit[i]);
            }
        }
    }

    printf("V matrix is:\n");
    for (int i=0;i<=n1;i++) {
        for (int j=0;j<=n2;j++)
        {
            printf("%d\t", v[i][j]);
        }
        printf("\n");
    }

    printf("If Selected Then 1 Else 0:\n");
    int i=n1,j=n2;
    while(i>0&&j>0)
    {
```

```
47              if(v[i][j]!=v[i-1][j])
48              {
49                  selected[i] = 1;
50                  j=j-weight[i];
51              }
52              i--;
53          }
54
55          printf("Selection array of profits (1 if selected, 0
        otherwise):\n");
56          for (int i = 1; i <= n1; i++)
57              printf("%d ", selected[i]);
58          printf("\n");
59      }
60
61      int main()
62      {
63          int profit[100],weight[100];
64          int num1,num2;
65          printf("Enter Number Of Obj: ");
66          scanf("%d",&num1);
67          for(int i=1;i<=num1;i++)
68          {
69              printf("Enter Profit: ");
70              scanf("%d",&profit[i]);
71              printf("Enter Weight: ");
72              scanf("%d",&weight[i]);
73          }
74          printf("\nEnter Max Capacity: ");
75          scanf("%d",&num2);
76          knapsack(num1,num2,profit,weight);
77          return 0;
78      }
```

Output:

```
Enter Number Of Obj: 4
Enter Profit: 3
Enter Weight: 4
Enter Profit: 2
Enter Weight: 4
Enter Profit: 4
Enter Weight: 4
Enter Profit: 4
Enter Weight: 4

Enter Max Capacity: 4
V matrix is:
0       0       0       0       0
0       0       0       0       3
0       0       0       0       3
0       0       0       0       4
0       0       0       0       4
If Selected Then 1 Else 0:
Selection array of profits (1 if selected, 0 otherwise):
0 0 1 0
PS C:\Users\KUSHAGRA\.vscode\sem 3>
```